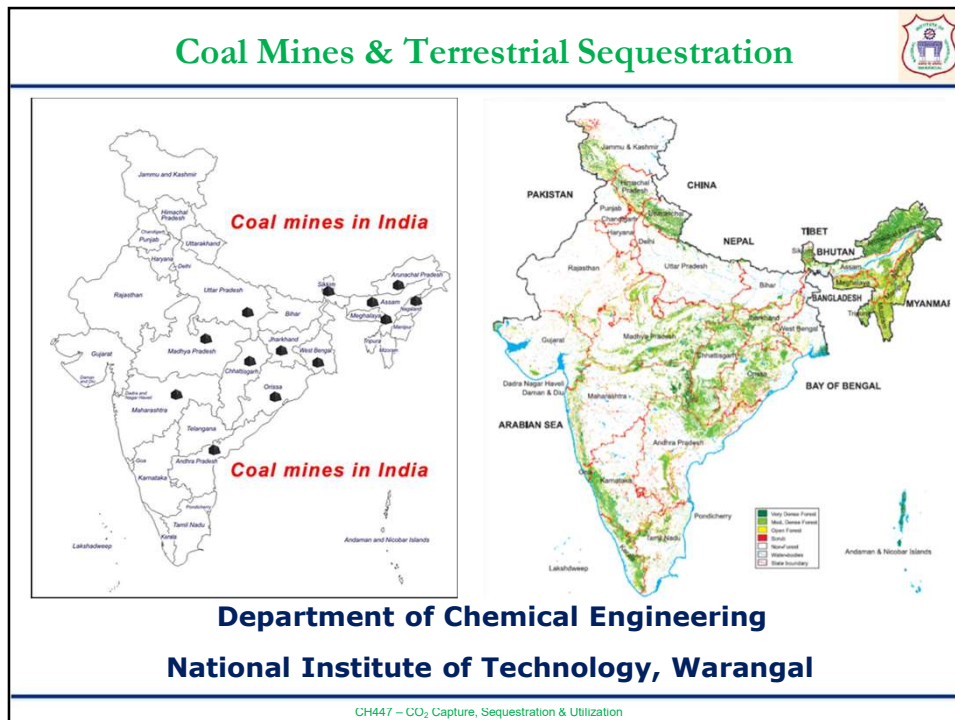




Carbon Capture Options

- Capturing CO₂ from the major stationary sources and its storage into deep geological formations is considered as a potential mitigation option.

Depleted Oil and Gas Reservoirs

Enhanced Oil Recovery

Enhanced Coal Bed Methane Recovery

Deep Unminable Coal Seams

Unused saline water-saturated formations

Geological sites – shales, basalt rocks, cavities

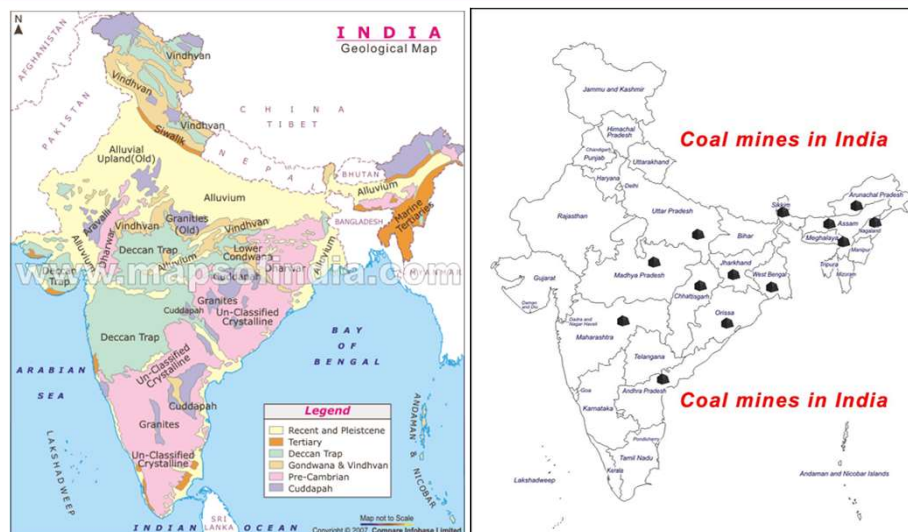
Oceans

CH447 – CO₂ Capture, Sequestration & Utilization

INDIA – Coal Production								
2022-2023								
COAL PRODUCTION AND OFFTAKE APRIL-AUG- 2022-2023 (IN MT)								
	Production*				Offtake/Dispatch*			
Company	Target	Ach. (Prov)	Actual During Cores period of previous year	Growth (%)	Target	Ach. (Prov.)	Actual During Cores period of previous year	Growth (%)
CIL	700.00	253.310	209.163	21.11	700.00	283.380	260.243	8.89
SCCL	70.00	24.305	25.401	-4.31	70.00	24.696	26.708	-7.53
Captive*	130.00	44.570	27.949	59.47	130.00	45.762	30.221	51.42
Others*	11.00	2.218	1.789	23.97	11.00	2.220	1.808	22.76
Total	911.00	324.403	264.302	22.74	911.00	356.058	318.980	11.62
Source: CCO report Production and dispatch target in respect of coal companies and others for Annual Plan 2022-23 and Annual Action Plan 2022-23								
Note: Production and Offtake of CIL is including Gare Palma IV/1 and IV 2&3 and production of captive is excluding these blocks.								
India has got large coal deposits https://coal.nic.in/content/production-and-supplies								
CH447 – CO ₂ Capture, Sequestration & Utilization								
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INDIA – Coal Mines Capacities				
(Million tonnes)				
State	Proved	Indicated	Inferred	Total
JHARKHAND	45563	31439	6150	83152
ODISHA	37391	34165	7739	79295
CHHATTISGARH	20428	34576	2202	57206
WEST BENGAL	14156	12869	4643	31667
MADHYA PRADESH	11958	12154	3875	27987
TELANGANA	10475	8576	2651	21702
MAHARASHTRA	7178	3074	2048	12299
ANDHRA PRADESH	0	1149	432	1581
BIHAR	161	813	392	1367
UTTAR PRADESH	884	178	0	1062
MEGHALAYA	89	17	471	576
ASSAM	465	57	3	525
NAGALAND	9	0	402	410
SIKKIM	0	58	43	101
ARUNACHAL PRADESH	31	40	19	90
Total	148787	139164	31069	319020
(Source: Geological Survey of India)				
https://coal.nic.in/content/production-and-supplies				
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INDIA – Coal Production



- Lower Gondwana coal reserves spread over 64000 km² area with 95% coals of INDIA.

<http://aksgb.blogspot.com/2016/09/coal-deposits-and-mines-in-india.html>

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INDIA – Coal Inventory

Table 1 : Depth-wise Gondwana coal resources of different states of India as on 1 April 2012

State	Resource estimate under depth			Total reserve (million Tonnes)
	0–300 m	300–600 m	600–1200 m	
Andhra Pradesh	9,654.17	8,754.74	3,745.95	22,154.86
Bihar	160.00	–	–	160.00
Chhattisgarh	37,755.15	11,609.08	1,481.92	50,846.15
Jharkhand	41,118.05	16,321.79	8,703.94	80,356.20**
**Jharia	-----14,212.42-----			
Maharashtra	7,522.32	3,165.28	194.49	1,0882.09
Madhya Pradesh	15,367.65	8,656.20	352.41	24,376.26
North-eastern states	1,394.66	202.00	–	1,596.66
Odisha	46,653.67	23,086.73	1,707.01	71,447.41
Uttar Pradesh	1,061.80	–	–	1,061.80
West Bengal	13,155.05	12,127.36	5,333.31	30,615.72
Total	173,842.52	83,923.18	21,519.03	293,497.15
**Jharia	-----14,212.42-----			
% share	59.23	28.60	7.33	100
**Jharia	-----4.84%-----			

**Jharia coalfield reserve position is available in group of 0–600 m depth range only. The reserve position within 300 m depth over is found in all the active coalfields except a few hidden pockets under basaltic flow or soil cover
Source: Compiled by authors.

- Inventory of coal is estimated from the data available from
1. Survey of India
 2. Central mine planning and design institute
 3. Mineral exploration corporation
 4. Singareni Collieries

- Most of the areas are not covered
1. Misleading surface exposures
 2. Terrain is unfriendly
 3. Difficulty in exploration

- This means lot of potential sites for CO₂ storage

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Coal Bed Charecteristics



- **Concealed Coal Fields** – These are under deep cover for which exact extension is just indicative by geophysical prospect and bore hole data from oil and gas exploration.
- Mining limit is decided by the quality, fuel value, market demand, market price, basin location, abundance of coal.
- **Unmienable coal beds** – a buffer zone of 100 m cap rock is suggested between the mineable seams and unmienable coal beds to contain the injected gas in coal beds.
- Storage characteristics in unmienable coal beds
 1. Possibility of CO₂ injection
 2. CH₄ drainage out of bed
 3. Gas sequestration potential
 4. Surface area of microporous system
 5. Thickness of coal seems
- CO₂/CH₄ selectivity in coal pores is 2 : 1

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Possible Storage Sites in INDIA



Identified candidates for CO₂ storage in India

Category of coalbeds	Grade of coal	Candidates/basins
Unmineable coalbeds in well-delineated coalfields Grey areas	Power grade coal	Singrauli, Mand Raigarh, Talcher, Godavari
	Coking coal	Jharia, East Bokaro, Sohagpur, S. Karanpura
	Superior non-coking coal	Raniganj, South Karanpura
Concealed coalfields	Power grade coal	Talcher
	Tertiary age coal	Cambay basin, Barmer Sanchor basin
	Power grade coal	West Bengal Gangetic plain, Birbhum, Domra Panagarh, Wardha valley extension, Kamptee basin extension

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Coal Bed Characteristics



- Without CH₄ gas information of coal bed empirical equations are used for gas storage estimation.
- Proximate analysis, vitrinite reflectance data will give gas recovery and storage capacity.

Proximate analysis and rank of unmineable and grey area coalbeds

Coalfields	Basic parameters (mmf* basis)		Vitrinite reflectance Av. Ro (%)
	VM (%)	FC (%)	
East Bokaro	28–36	85–90	0.85–1.05
South Karanpura	37–40	80–84	0.60–0.80
Jharia-Barakar	17–35	87–93	0.90–1.30
Raniganj	39–44	79–90	0.70–0.85
Rajmahal-Barakar	38–40	78–81	0.45–0.50
Singrauli-Barakar	37–45	78–81	0.45–0.50
Sohagpur	34–40	80–87	0.55–0.65
Pench valley	32–40	82–89	0.50–0.60
Wardha valley	35–40	78–82	0.55–0.60
Godavari valley	35–42	78–83	0.55–0.60
Talcher	35–45	79–82	0.50–0.55

*mmf (mineral matter free basis)

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Coal Bed Possible Storage



- Unmineable coal beds – Possible options for CO₂ storage irreversibly.

Unmineable coal reserve and CO₂ storage capacity

Coalfield	Estimated adsorption capacity of CO ₂ (m ³ /t)	Coal reserve (Mt)	CO ₂ storage capacity (Bm ³)	CO ₂ storage capacity (Mt)	CO ₂ storage capacity (90%) (Mt)
Singrauli	Average 20.0	37.0	0.74	1.46	1.32
Mand Raigarh	Range 16.0–23.0 Average 19.0	79.0	1.50	2.97	2.67
Talcher	Range 17.2–24.8 Average 20.4	1,017.0	20.80	41.18	37.06
Godavari	Range 16.8–22.2 Average 19.2	1,976.0	38.02	75.28	67.75

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Coal Bed Possible Storage							
➤ Grey Coal Reserves – below 1200 m depth cover							
Estimated CO ₂ adsorption capacity in grey area coalbeds							
Coalfield	Estimated CO ₂ adsorption capacity (m ³ /t)	Cumulative coal seam thickness (m)	Block area (km ²)	Coal reserve (Bt)	CO ₂ storage capacity (Bm ³)	CO ₂ storage capacity (Mt)	CO ₂ storage capacity (90%) (Mt)
South Karanpura	Range 19.5–28.0 Average 24.5	73.0	76.0	0.75	18.35	36.33	21.80
East Bokaro	Range 22.3–33.5 Average 28.1	100.0	113.0	1.53	42.90	84.94	76.45
Jharia	Range 22.0–56.0 Average 34.5	40.0	193.0	1.04	35.96	71.20	64.08
Raniganj	Range 20.8–29.0 Average 24.0	30.0	240.0	0.97	23.33	46.19	41.57
Sohagpur	Range 18.9–26.4 Average 22.6	15.0	450.0	0.91	20.59	40.76	36.69
Talcher	Range 17.2–24.8 Average 20.4	120.0	149.0	2.41	49.24	97.49	87.75

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Coal Bed Possible Storage							
➤ Concealed Coal Reserves – deep coal seams or reserves							
Concealed area coal reserve and CO ₂ storage capacity							
Coalfield	Estimated adsorption capacity (m ³ /t)	Cumulative thickness of the coal seams (m)	Area of the block (km ²)	Coal reserve (Bt)	CO ₂ storage capacity (Bm ³)	CO ₂ storage capacity (Mt)	CO ₂ storage capacity (90%) (Mt)
Cambay basin	Range 13.8–19.6 Average 16.7	102.0	6,900	63.0	1,057.81	2,094.45	1,885.02
Barmer Sanchor basin	Range 128–18.4 Average 15.6	100.0	6,700	60.0	936.00	1,853.28	1,667.95
West Bengal Gangetic plain	Range 16.4–23.2 Average 18.3	–	–	7.2	131.76	260.88	234.80
Birbhum coalfield	Range 17.2–24.8 Average 20.2	100.0	312.0	4.2	85.08	168.46	151.61
Domra Panagarh	Range 18.6–25.8 Average 21.8	48.0	116.0	0.751	16.39	32.45	29.20
Wardha valley extension	Range 15.7–22.8 Average 17.8	13.0	212.0	0.37	6.62	13.11	11.80
Kamptee extension	Range 7.2–9.2 Average 8.1	14.0	300	0.57	9.81	19.42	17.48

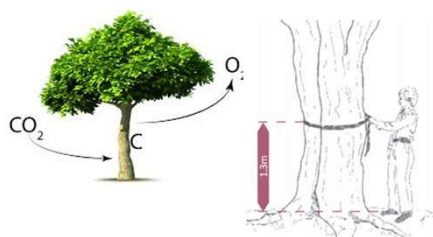
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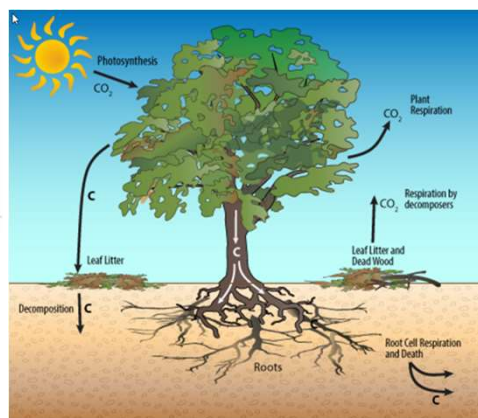
Terrestrial Sequestration



- Forests are one of the natural sinks of CO_2 .
- Trees convert the CO_2 into biomass by photosynthesis and release same CO_2 to environment through respiration and decomposition.



Circumference (cm)	Tree dry weight (kg)
50	106
100	668
150	1,964
200	4,221
225	5,771
250	7,641
275	9,842
300	12,410
325	15,350
350	18,700
400	26,674



<https://rgsgeogy.wordpress.com/mrcs-tree-carbon-content-calculator/>

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Forest Carbon Pools

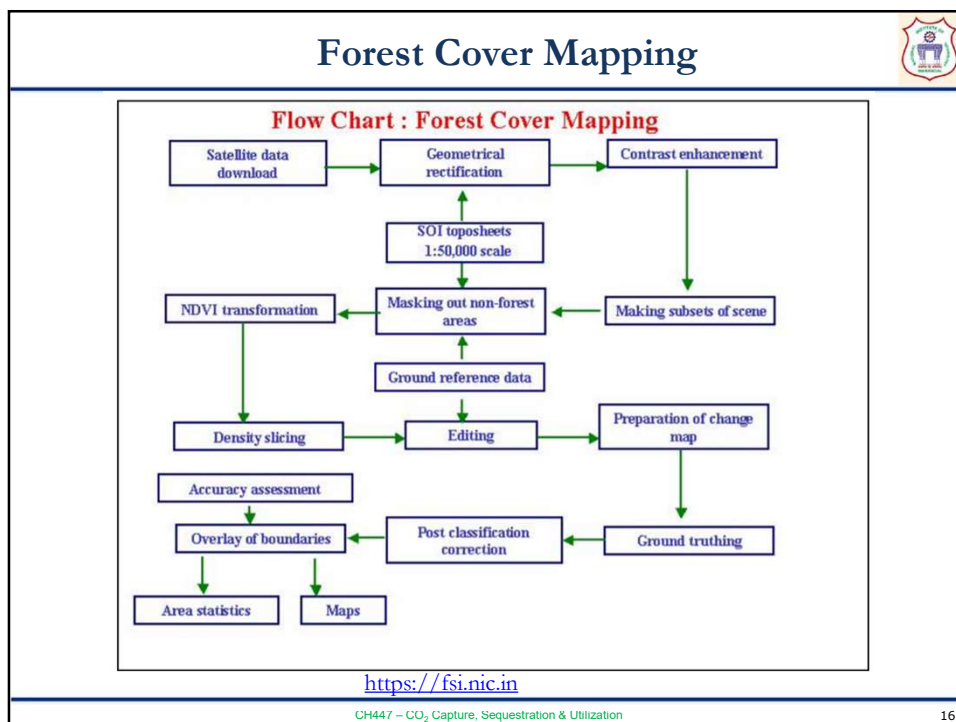
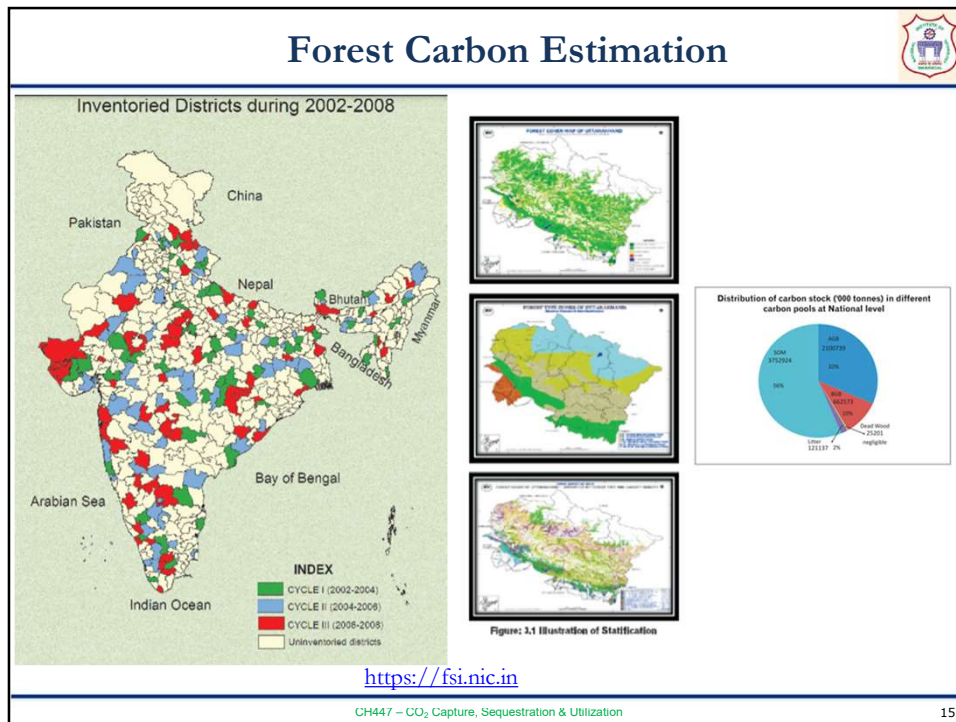


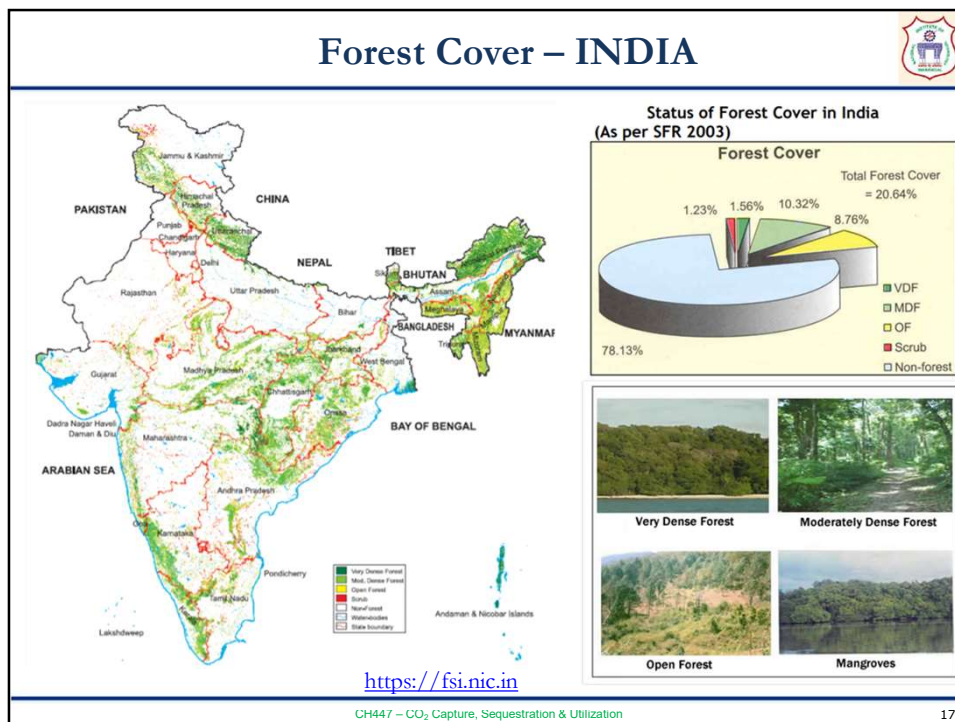
Pools		Description
Living Biomass	Above ground biomass (AGB)	All living biomass above the soil including stem, stump, branches, bark, seeds and foliage.
	Below ground biomass (BGB)	All living biomass of live roots. Fine roots of less than 2mm diameter (country specific) are often excluded because these often cannot be distinguished empirically from soil organic matter or litter.
Dead Organic Matter	Dead wood	Includes all non living woody biomass not contained in the litter, either standing or lying on the ground. Dead wood also includes dead roots and stumps larger than or equal to 10cm in diameter or any other diameter used by the country.
	Litter	Includes all non-living biomass with a diameter less than a minimum diameter chosen by the country (for FSI 5 cm), lying dead, in various states of decomposition above the mineral or organic soil.
Soil	Soil organic matter	Includes organic carbon in mineral and organic soils (including peat) to a specific depth chosen by the country (for FSI 30 cm) and applied consistently through the time series.

https://fsi.nic.in/carbon_stock/chapter-2.pdf

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Forest Carbon Estimation

Forest Types for stratification

FTM Code	FTM Name
1	Tropical Wet Evergreen Forests - North East
2	Tropical Wet Evergreen Forests - Western Ghats
3	Tropical Semi Evergreen Forests - North East
4	Tropical Semi Evergreen Forests - Eastern Deccan
5	Tropical Semi Evergreen Forests - Western Ghats
6	Tropical Moist Deciduous Forests
7	Littoral & Swamp Forests
8	Tropical Dry Deciduous Forests
9	Tropical Thorn Forests
10	Tropical, Subtropical Dry Evergreen & Broadleaved Hill Forests
11	Subtropical Pine Forests
12	Montane & Moist Temperate Forests
13	Sub Alpine & Dry Temperate Forests
14	Alpine Scrub
15	Plantation/TOF

<https://fsi.nic.in>

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Forest Carbon Estimation



Forest Type and Density wise Carbon Stock in different carbon pools

(000 tonnes)																	
Forest Type Stratum	Density	Area in Kn ²	AGB	BGB	Dead wood	Litter	SOM	Total	Forest Type Stratum	Density	Area in Kn ²	AGB	BGB	Dead wood	Litter	SOM	Total
Tropical Wet Evergreen Forests - North East	VDF	883	4,353	1,506	106	239	10,038	16,242	Tropical Dry Deciduous Forests	VDF	13,434	63,200	32,669	615	7,888	79,040	2,03,413
Tropical Wet Evergreen Forests - North East	MDF	3,343	9,175	2,731	122	1,062	24,357	37,447	Tropical Dry Deciduous Forests	MDF	1,30,339	7,71,107	3,02,783	1,822	7,836	6,92,382	17,76,030
Tropical Wet Evergreen Forests - North East	OF	1,763	3,211	1,277	35	504	11,747	16,774	Tropical Dry Deciduous Forests	OF	1,27,519	1,63,454	64,182	1,677	5,249	3,81,376	6,15,938
Tropical Wet Evergreen Forests - Western Ghats	VDF	3,596	25,412	8,790	1,474	2,582	34,381	72,640	Tropical Thorn Forests	VDF	103	142	56	4	22	302	526
Tropical Wet Evergreen Forests - Western Ghats	MDF	9,239	43,067	14,898	324	3,240	78,072	1,39,601	Tropical Thorn Forests	MDF	3,079	4,067	1,597	76	445	11,480	17,666
Tropical Wet Evergreen Forests - Western Ghats	OF	1,747	2,082	720	46	396	14,240	17,485	Tropical Thorn Forests	OF	8,568	4,270	1,677	107	650	17,565	24,269
Tropical Semi Evergreen Forests - Western Ghats	VDF	11,743	68,983	14,188	1,600	3,198	1,57,270	2,45,240	Tropical, Subtropical Dry Evergreen & Broadleaved Hill Forests	VDF	1,524	5,362	2,105	139	239	14,097	21,941
Tropical Semi Evergreen Forests	MDF	38,328	1,15,260	23,707	1,402	9,311	3,30,093	4,78,773	Tropical Subtropical Dry Evergreen & Broadleaved Hill Forests	MDF	8,535	21,552	8,463	312	2,049	69,380	1,01,756
Tropical Semi Evergreen Forests	OF	32,416	30,078	6,186	650	5,812	1,40,071	1,82,797	Tropical Subtropical Dry Evergreen & Broadleaved Hill Forests	OF	10,084	16,465	6,465	202	1,805	69,816	94,753
Tropical Semi Evergreen Forests - Eastern Deccan	VDF	4	15	3	1	15	33	67	Subtropical Pine Forests	VDF	1,116	5,620	1,419	133	164	10,175	17,511
Tropical Semi Evergreen Forests - Eastern Deccan	MDF	186	754	155	7	320	952	2,187	Subtropical Pine Forests	MDF	10,235	32,462	8,194	375	1,322	64,629	1,06,981
Tropical Semi Evergreen Forests - Eastern Deccan	OF	201	511	105	5	287	697	1,605	Subtropical Pine Forests	OF	7,030	13,837	3,493	141	676	36,412	54,559
Tropical Semi Evergreen Forests - Western Ghats	VDF	1,561	7,789	1,602	615	1,228	16,016	27,250	Montane & Moist Temperate Forests	VDF	4,782	30,649	7,736	652	1,537	43,803	84,376
Tropical Semi Evergreen Forests - Western Ghats	MDF	10,214	35,286	7,258	359	7,443	69,901	1,20,246	Montane & Moist Temperate Forests	MDF	16,603	83,972	21,197	608	4,056	1,22,207	2,32,040
Tropical Semi Evergreen Forests - Western Ghats	OF	3,902	4,716	970	103	2,670	20,330	29,300	Montane & Moist Temperate Forests	OF	11,559	28,666	7,236	232	2,243	64,338	1,02,715
Tropical Moist Deciduous Forests	VDF	10,521	39,096	8,041	1,232	2,888	80,244	1,31,501	Sub Alpine & Dry Temperate Forests	VDF	2,012	15,907	4,313	241	415	20,077	40,952
Tropical Moist Deciduous Forests	MDF	78,237	1,99,186	40,968	6,448	25,853	4,73,833	7,46,288	Sub Alpine & Dry Temperate Forests	MDF	9,249	42,414	11,499	338	1,030	60,809	1,16,091
Tropical Moist Deciduous Forests	OF	51,430	60,728	12,491	2,056	9,435	2,50,881	3,35,591	Sub Alpine & Dry Temperate Forests	OF	7,040	23,162	6,280	141	504	31,004	61,091
Littoral & Swamp Forests	VDF	1,190	8,997	3,112	3	151	9,848	22,111	Alpine Scrub	VDF	1,615	13,187	3,575	193	204	13,961	31,120
Littoral & Swamp Forests	MDF	1,969	8,456	2,925	4	134	11,351	22,870	Alpine Scrub	MDF	1,312	3,991	1,082	48	139	10,204	15,464
Littoral & Swamp Forests	OF	1,708	2,396	825	4	87	5,624	6,907	Alpine Scrub	OF	1,194	2,700	732	24	98	4,745	8,299
									Plantation/TOF	VDF	545	1,723	354	3	121	5,170	7,372
									Plantation/TOF	MDF	11,781	35,025	7,197	421	2,835	91,035	1,36,513
									Plantation/TOF	OF	23,712	29,254	5,811	0	2,774	98,539	1,35,188
									Total		6,77,888	21,86,738	6,62,973	25,281	1,21,137	37,52,824	66,62,974

<https://fsi.nic.in>

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Forest Carbon Estimation



Change in carbon stock in forest land between 1994 and 2004.

(Million tonnes)

Component	Carbon Stock in forest land in 1994	Carbon stock in forest land in 2004	Net change in Carbon stock
Above Ground Biomass	1784	2101	317
Below Ground Biomass	563	663	100
Dead wood	19	25	6
Litter	104	121	17
Soil	3601	3753	152
Total	6071	6663	592

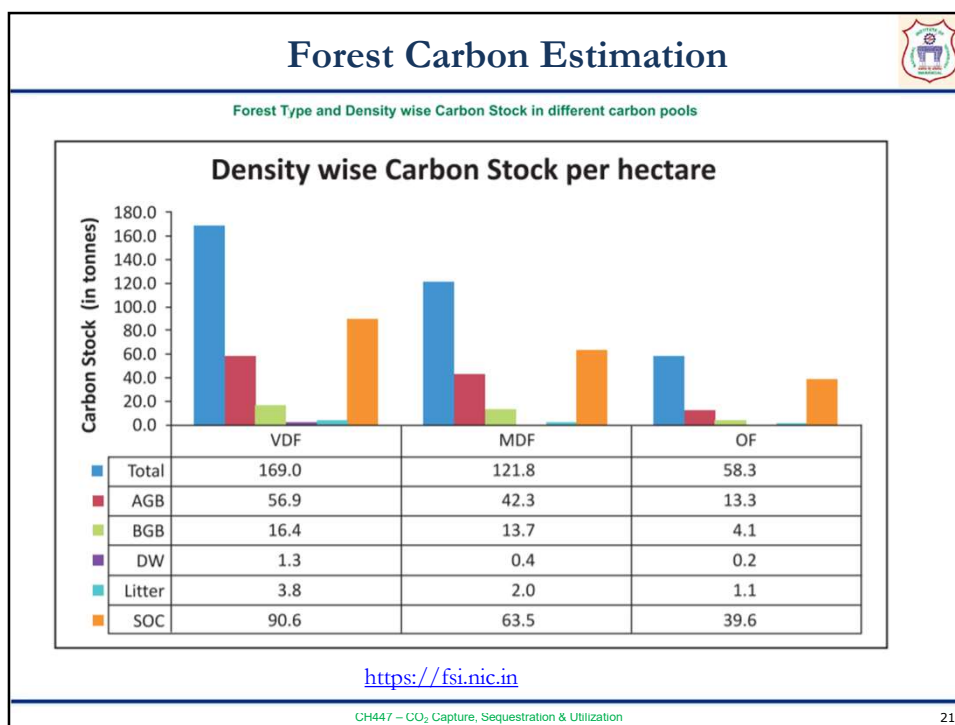
The total estimated carbon stored in harvested wood products is 257 million tonnes.

Items	Household	Commercial Units		Total
		Own Account Establishments	Other Establishments	
Volume of wood (m cu m)	652.99	50.25	155.64	858.88
Biomass (m tons)	424.44	32.66	101.17	558.27
Carbon (m tons)	195.24	15.02	46.54	256.80

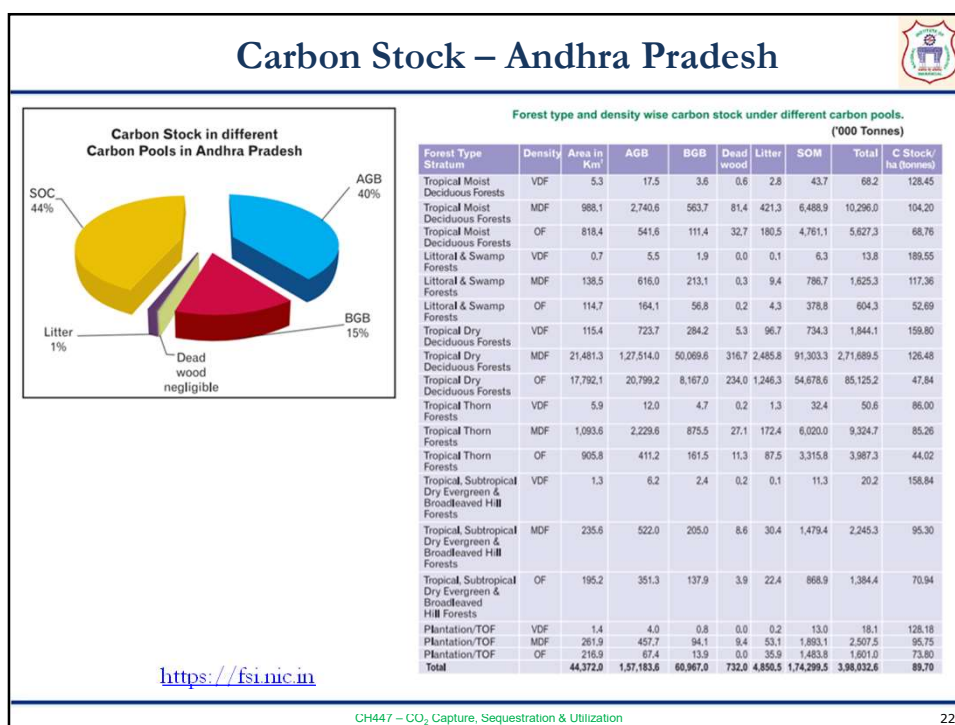
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CH447 – CO₂ Capture, Sequestration & Utilization

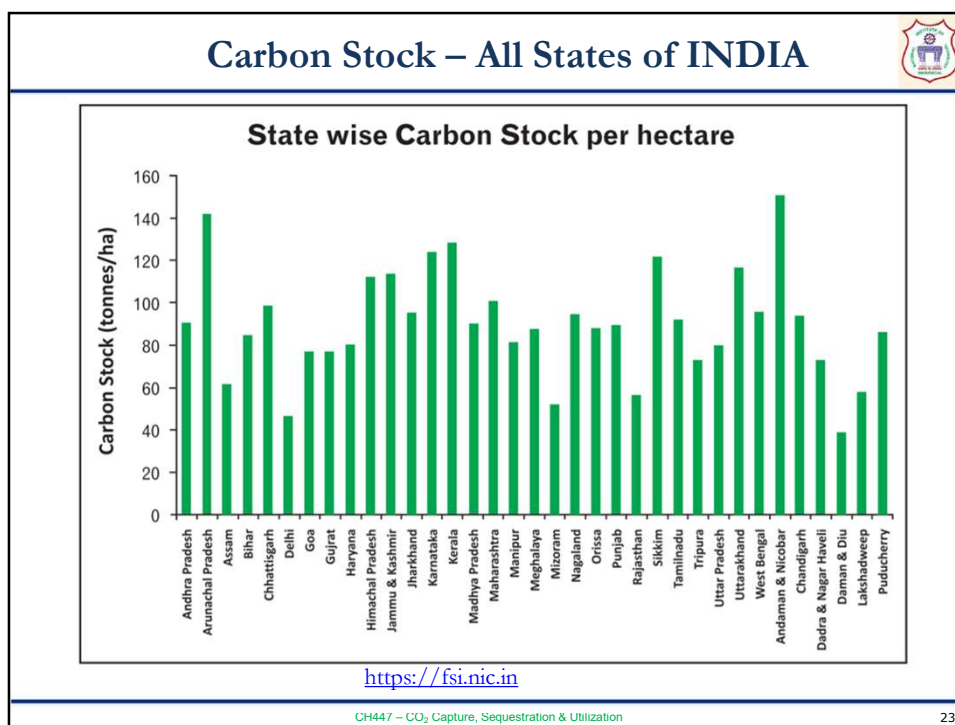
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Forest Cover of NE States

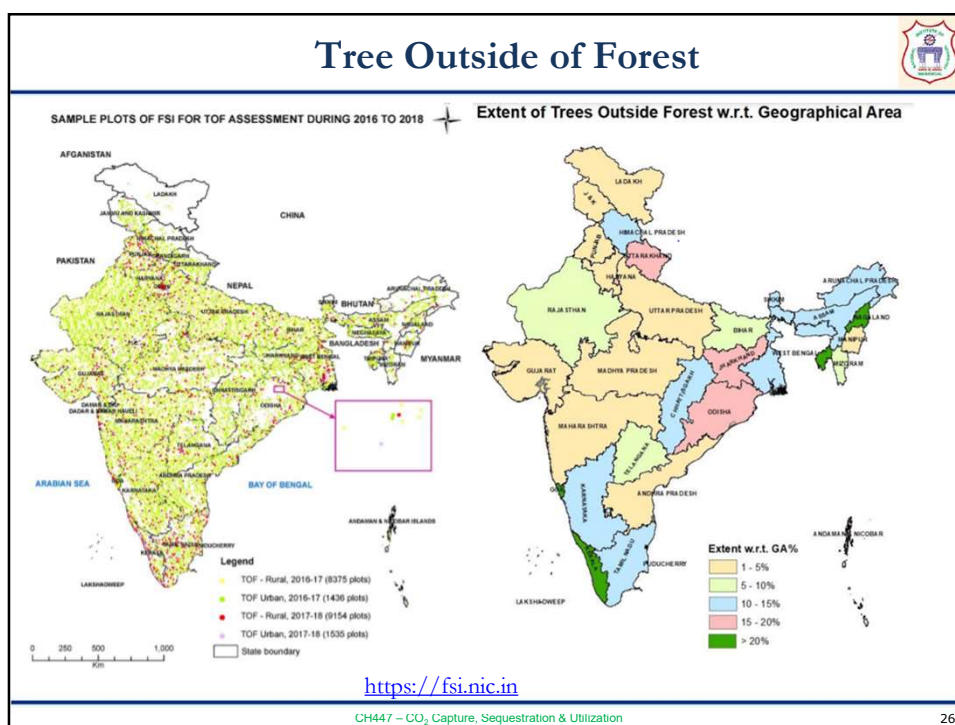
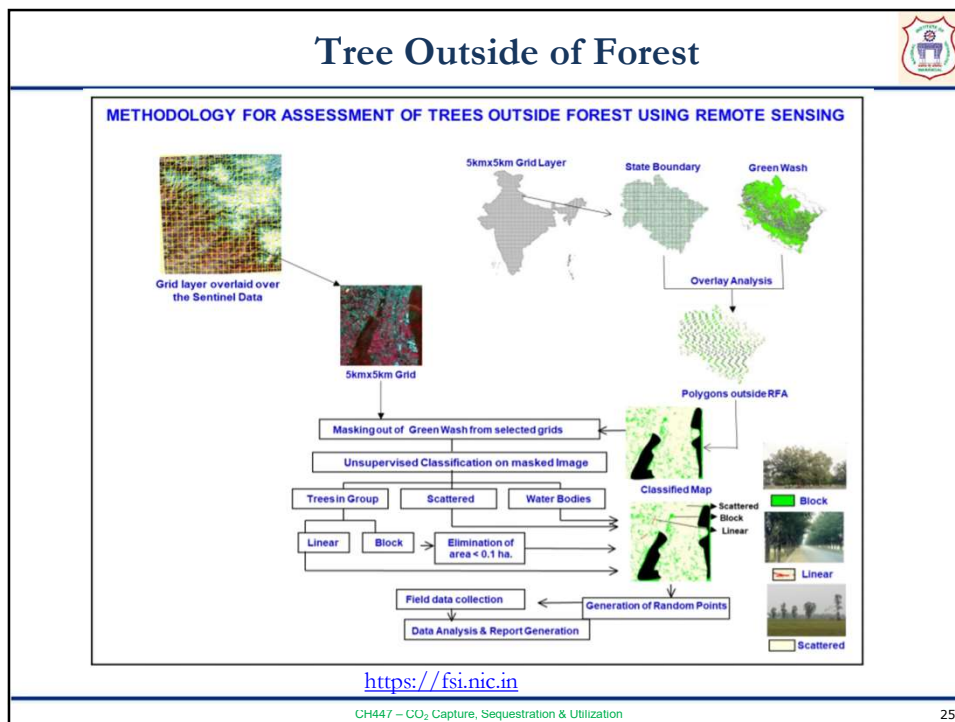
Forest cover in the north-eastern states in 2009

State/UT	Geographical area (GA)	Forest cover				% of GA	Change*	Scrub
		Very dense forest	Moderate dense forest	Open forest	Total			
Arunachal Pradesh	83,743	20,868	31,519	15,023	67,410	80.50	– 74	122
Assam	78,438	1,444	11,404	14,825	27,673	35.28	– 19	182
Manipur	22,327	730	6,151	10,209	17,090	76.54	– 190	1
Meghalaya	22,429	433	9,775	7,067	17,275	77.02	– 46	485
Mizoram	21,081	134	6,086	12,897	19,117	90.68	– 66	1
Nagaland	16,579	1,293	4,931	7,094	13,318	80.33	– 146	3
Sikkim	7,096	500	2,161	698	3,359	47.34	0	363
Tripura	10,486	109	4,686	3,182	7,977	76.04	– 8	72
Grand total	262,179	25,511	76,713	70,995	173,219	66.07	– 549	1,229

(Area in km²)

CH447 – CO₂ Capture, Sequestration & Utilization

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Tree Outside of Forest



North-eastern state-wise growing stock in forests and TOF (2010)

State	Geographical area (km ²)	Recorded forest area (km ²)	Volume of growing stock (Mm ³)		
			In forest	In TOF	Total
Arunachal Pradesh	83,743	51,540	492.689	74.516	567.205
Assam	78,438	26,832	173.494	41.336	214.830
Manipur	22,327	17,418	70.878	10.691	81.569
Meghalaya	22,429	9,496	45.411	20.964	66.375
Mizoram	21,081	16,717	68.042	9.392	77.434
Nagaland	16,579	9,222	40.955	12.681	53.636
Sikkim	7,096	5,841	18.832	2.017	20.849
Tripura	10,486	6,294	21.864	7.391	29.255
Total of the states of North-eastern India	262,179	143,360	932.165	178.988	1,111.153
Total of all states or UT of India	32,87,263	769,538	4,498.731	1,548.427	6,047.158
Percentage of north-eastern India	7.976	18.629	20.721	11.559	18.375

Extent of Trees Outside Forest (ISFR 2019)

State/UTs	TGA (sq km)	Tree cover 2019 (sq km) (A)	Forest cover outside RFA 2019 (sq km) (B)	Extent of TOF 2019 (sq km) (A+B)	% of Geographical Area of the State/UT
Andhra Pradesh	162968	3,914	5,018	8,932	5.48
Assam	83743	648	7,967	8,615	10.53
Bihar	94163	2,003	2,537	4,540	4.82
Chhattisgarh	135192	4,346	13,195	17,541	12.90
Dadra & Nagar Haveli	1483	129	136.37	265	17.92
Goa	3702	272	1,063	1,335	36.05
Gujarat	196244	6,912	5,072	11,984	6.11
Haryana	44212	1,565	1,229	2,794	6.32
Himachal Pradesh	55673	839	4,796	5,635	10.10
J&K and Ladakh*	222236	7,844	11,396	19,240	8.70
Jharkhand	79716	2,657	11,402	14,059	17.64
Karnataka	191791	6,257	16,104	22,361	11.66
Kerala	38852	2,936	11,507	14,443	37.17
Madhya Pradesh	308252	8,339	12,730	21,069	6.83
Maharashtra	307715	10,806	16,739	27,545	9.76
Manipur	22327	173	1,829	2,002	8.96
Meghalaya	22429	710	2,275	2,985	13.31
Mizoram	21081	441	270	711	3.37
Nagaland	16579	362	3,758	4,121	24.86
Odisha	158707	4,642	18,510	23,152	15.26
Punjab	50582	1,562	1,065	2,627	5.28
Rajasthan	342239	8,112	4,348	12,460	3.64
Sikkim	7096	36	996	1,032	14.54
Tamil Nadu	130060	4,830	8,775	13,605	10.46
Telangana	112077	2,514	2,313	4,827	4.31
Tripura	10486	231	2,275	2,506	23.90
Uttar Pradesh	240928	7,342	8,611	15,953	6.58
Uttarakhand	53485	941	7,513	8,454	15.62
West Bengal	88752	2,006	9,825	11,831	13.33
Andaman & Nicobar Islands	8249	41	521	562	6.81
Chandigarh	114	25	13.76	39	34.41
Dadra & Nagar Haveli	491	28	47	75	15.32
Daman & Diu	111	5	20.49	25	22.52
Lakshadweep	30	0.29	27.10	27	91.30
Pondicherry	490	23	51.41	74	15.15
Total	3287469	94027	198813	293840	8.94

* includes J&K and Ladakh area outside LOC that is under illegal occupation of Pakistan and China

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